

You have 2 hours for completing the exam.

No electronic devices are allowed (i.e. calculators, mobile phones, etc.).

All answers must include the supporting deductions and calculations (if the calculations are too heavy then they can be left implied in the final answer).

If some answer does not include all the necessary information, it will be deemed invalid.

Name and surname:

1. (2 points) Solve the following recurrence equation by means of generating functions

$$a_n = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n = 1, \\ a_{n-1} + 2a_{n-2} & \text{for } n \geq 2. \end{cases}$$

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The recurrence can be reformulated as

$$a_{n+2} = a_{n+1} + 2a_n.$$

We multiply both sides of the equation by z^n and sum over $n \geq 0$:

$$\begin{aligned} \sum_{n=0}^{\infty} a_{n+2} z^n &= \sum_{n=0}^{\infty} a_{n+1} z^n + \sum_{n=0}^{\infty} a_n z^n \\ \frac{A(z) - a_0 - a_1 z}{z^2} &= \frac{A(z) - a_0}{z} + 2A(z) \\ \frac{A(z) - 1}{z^2} &= \frac{A(z) - 1}{z} + 2A(z) \end{aligned}$$

By solving for $A(z)$ we get

$$A(z) = \frac{1-z}{1-z-2z^2} = \frac{1-z}{(1+z)(1-2z)}.$$

Now we decompose $A(z)$ as a sum of partial fractions:

$$\begin{aligned} A(z) &= \frac{c_1}{1+z} + \frac{c_2}{1-2z} \\ &= \frac{2}{3(1+z)} + \frac{1}{3(1-2z)}. \end{aligned}$$

Finally, with the aid of the table at the back of the exam we get

$$A(z) = \frac{2}{3} \sum_{n=0}^{\infty} (-1)^n z^n + \frac{1}{3} \sum_{n=0}^{\infty} 2^n z^n,$$

$$a_n = \frac{2}{3}(-1)^n + \frac{1}{3}2^n.$$

2. Let $\Omega = \{a, b, c, d\}$ and $\mathcal{F} = \{\emptyset, \{a\}, \{b, c\}, \{d\}, \{a, b, c\}, \{b, c, d\}, \{a, d\}, \Omega\}$.

(a) (1 point) Prove that $(\Omega, \mathcal{F})$ is a measurable space.

(b) (1 point) Let $\mathbb{P} : \mathcal{F} \rightarrow [0; 1]$ be defined as follows: $\mathbb{P}(\{a\}) = \frac{1}{3}$, $\mathbb{P}(\{b, c\}) = \frac{1}{4}$ and $\mathbb{P}(\{d\}) = \alpha$. Determine the value of α so that $\mathbb{P}$ is a probability measure over $(\Omega, \mathcal{F})$.

(c) (1 point) Calculate $\mathbb{P}(\{b, c, d\})$ and $\mathbb{P}(\{a, d\})$.

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(a) In order to prove that $(\Omega, \mathcal{F})$ is a measurable space we just have to show that $\mathcal{F}$ is a σ -algebra over Ω :

i. $\Omega \in \mathcal{F}$ ✓

ii. For all $A \in \mathcal{F}$ we have $\bar{A} \in \mathcal{F}$:

- $\bar{\Omega} = \emptyset \in \mathcal{F}$ ✓
- $\overline{\{a\}} = \{b, c, d\} \in \mathcal{F}$ ✓
- $\overline{\{b, c\}} = \{a, d\} \in \mathcal{F}$ ✓
- $\overline{\{d\}} = \{a, b, c\} \in \mathcal{F}$ ✓

iii. If $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$. Obviously, we don't have

to check all the finite or infinite unions that can be constructed with the elements of $\mathcal{F}$. The following two-set combinations will suffice:

- $\{a\} \cup \{b, c\} = \{a, b, c\} \in \mathcal{F}$ ✓
- $\{a\} \cup \{d\} = \{a, d\} \in \mathcal{F}$ ✓
- $\{b, c\} \cup \{d\} = \{b, c, d\} \in \mathcal{F}$ ✓
- $\{a\} \cup \{b, c, d\} = \{a, b, c, d\} \in \mathcal{F}$ ✓
- $\{a, b, c\} \cup \{d\} = \{a, b, c, d\} \in \mathcal{F}$ ✓
- $\{a, d\} \cup \{b, c\} = \{a, b, c, d\} \in \mathcal{F}$ ✓
- $\{a, d\} \cup \{a, b, c\} = \{a, b, c, d\} \in \mathcal{F}$ ✓

- $\{a, d\} \cup \{b, c, d\} = \{a, b, c, d\} \in \mathcal{F}$ ✓
- $\{a, b, c\} \cup \{b, c, d\} = \{a, b, c, d\} \in \mathcal{F}$ ✓

Note that we have omitted the combinations where one set is a subset of the other.

(b) For $\mathbb{P}$ to be a probability measure over $(\Omega, \mathcal{F})$ we must have

$$\begin{aligned}\mathbb{P}(\Omega) &= \mathbb{P}(\{a\} \cup \{b, c\} \cup \{d\}) = \mathbb{P}(\{a\}) + \mathbb{P}(\{b, c\}) + \mathbb{P}(\{d\}) \\ &= \frac{1}{3} + \frac{1}{4} + \alpha = \frac{7}{12} + \alpha = 1,\end{aligned}$$

hence

$$\alpha = 1 - \frac{7}{12} = \frac{5}{12}.$$

However, we're not done yet, since we still have to verify that $\mathbb{P}$ is indeed a probability measure over $(\Omega, \mathcal{F})$, or in other words, that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. In order to do that we have to check that

- $\mathbb{P}(\Omega) = 1$ ✓
- For all $A \in \mathcal{F}$ we have $\mathbb{P}(A) \geq 0$ ✓
- If $A_1, A_2, \dots$ are **disjoint** subsets $\in \mathcal{F}$ then $\mathbb{P}\left(\bigcup_{i=0}^{\infty} A_i\right) = \sum_{i=0}^{\infty} \mathbb{P}(A_i)$.

So we have to verify the following combinations:

- $\mathbb{P}\left(\{a\} \cup \{b, c\}\right) = \mathbb{P}\left(\{a\}\right) + \mathbb{P}\left(\{b, c\}\right) = \frac{1}{3} + \frac{1}{4} = \frac{7}{12} = \mathbb{P}\left(\{a, b, c\}\right)$ ✓
- $\mathbb{P}\left(\{a\} \cup \{d\}\right) = \mathbb{P}\left(\{a\}\right) + \mathbb{P}\left(\{d\}\right) = \frac{1}{3} + \frac{5}{12} = \frac{3}{4} = \mathbb{P}\left(\{a, d\}\right)$ ✓
- $\mathbb{P}\left(\{b, c\} \cup \{d\}\right) = \mathbb{P}\left(\{b, c\}\right) + \mathbb{P}\left(\{d\}\right) = \frac{1}{4} + \frac{5}{12} = \frac{2}{3} = \mathbb{P}\left(\{b, c, d\}\right)$ ✓
- $\mathbb{P}\left(\{a, b, c\} \cup \{d\}\right) = \mathbb{P}\left(\{a, b, c\}\right) + \mathbb{P}\left(\{d\}\right) = \frac{7}{12} + \frac{5}{12} = 1 = \mathbb{P}\left(\{a, b, c, d\}\right)$ ✓
- $\mathbb{P}\left(\{a\} \cup \{b, c, d\}\right) = \mathbb{P}\left(\{a\}\right) + \mathbb{P}\left(\{b, c, d\}\right) = \frac{1}{3} + \frac{2}{3} = 1 = \mathbb{P}\left(\{a, b, c, d\}\right)$ ✓

- $\mathbb{P}\left(\{a, d\} \cup \{b, c\}\right) = \mathbb{P}\left(\{a, d\}\right) + \mathbb{P}\left(\{b, c\}\right) = \frac{3}{4} + \frac{1}{4} = 1 = \mathbb{P}\left(\{a, b, c, d\}\right) \quad \checkmark$
- $\mathbb{P}\left(\{a\} \cup \{b, c\} \cup \{d\}\right) = \mathbb{P}\left(\{a\}\right) + \mathbb{P}\left(\{b, c\}\right) + \mathbb{P}\left(\{d\}\right) = \frac{1}{3} + \frac{1}{4} + \frac{5}{12} = 1 = \mathbb{P}\left(\{a, b, c, d\}\right) \quad \checkmark$

Note: In principle the last line was not necessary because it was already verified when we calculated $\mathbb{P}(\{d\})$.

- (c) Both $\mathbb{P}\left(\{a, d\}\right)$ and $\mathbb{P}\left(\{b, c, d\}\right)$ were calculated above. Their respective values are $\frac{3}{4}$ and $\frac{2}{3}$.
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3. (2 points) Suppose that the floor of our house is covered with rectangular tiles of size $2a \times a$, as in Figure 1. If we flip a coin of radius 1 on the floor, there is a probability p that the coin **does not** intersect any of the lines. What should be the value of a so that $p > \frac{1}{2}$?

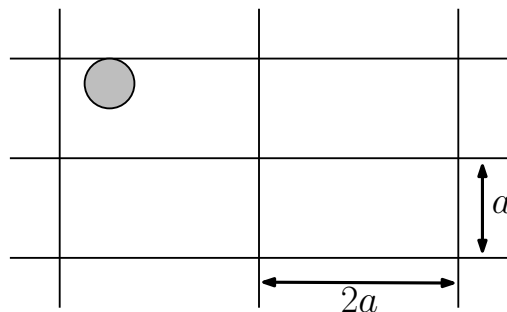


Figure 1: Tiled floor and coin of radius 1

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For the coin **not** to touch any of the lines, its center must fall inside the inner (dashed) rectangle, which we will call the *admissible region* (see Figure 2). Then, the probability that the center of the coin falls within the

admissible region is the area of the admissible region divided by the area of the outer rectangle, i.e.

$$p = \frac{(a-2)(2a-2)}{2a^2} = \frac{2a^2 - 6a + 4}{2a^2}.$$

So we have to solve the inequality

$$\begin{aligned} p = \frac{2a^2 - 6a + 4}{2a^2} &> \frac{1}{2} \\ 2a^2 - 6a + 4 &> \frac{2a^2}{2} \\ a^2 - 6a + 4 &> 0 \end{aligned}$$

The solutions to the latter inequality are $a < 3 - \sqrt{5}$ or $a > 3 + \sqrt{5}$, but the only one that makes sense is $a > 3 + \sqrt{5}$, since $3 - \sqrt{5} < 1$ (the radius of the coin).

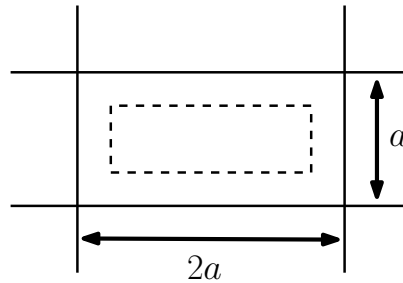


Figure 2: Admissible region for the center of the coin

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4. We have two urns, the first one containing 4 black balls and 2 white balls, while the second box contains 3 white balls and 2 black balls. An experiment consists of selecting a box at random, and then extracting two balls at once from the chosen box, also at random.
- (1 point) Calculate the probability that at least one of the extracted balls is white.
 - (2 points) Suppose that we don't know which box was chosen. If the two balls turn out to be white, calculate the probability that the first box was chosen.

Solució

- (a) Let U_i denote the event that the urn U_i is chosen (where $i \in \{1, 2\}$). Additionally, let W_1 be the event that **at least** one of the balls is white, and let B denote the event that **both** balls are black. If we want to calculate $\mathbb{P}(W_1)$ directly, we have to use the formula for the total probability, i.e.

$$\mathbb{P}(W_1) = \mathbb{P}(W_1|U_1) \mathbb{P}(U_1) + \mathbb{P}(W_1|U_2) \mathbb{P}(U_2)$$

Now, the easiest way to calculate $\mathbb{P}(W_1)$ is indirectly, by noting that $\mathbb{P}(W_1) = 1 - \mathbb{P}(B)$. Then

$$\begin{aligned} \mathbb{P}(B) &= \mathbb{P}(B|U_1) \mathbb{P}(U_1) + \mathbb{P}(B|U_2) \mathbb{P}(U_2) \\ &= \frac{\binom{4}{2}}{\binom{6}{2}} \frac{1}{2} + \frac{\binom{2}{2}}{\binom{5}{2}} \frac{1}{2} = \frac{6}{15} \frac{1}{2} + \frac{1}{10} \frac{1}{2} = \frac{1}{4} \end{aligned}$$

$$\text{Hence } \mathbb{P}(W_1) = 1 - \mathbb{P}(B) = 1 - \frac{1}{4} = \frac{3}{4}.$$

- (b) We now want to calculate $\mathbb{P}(U_1|W_2)$ where W_2 denotes the event that **both** balls are white. Let us first calculate $\mathbb{P}(W_2)$ by the formula of total probability:

$$\begin{aligned} \mathbb{P}(W_2) &= \mathbb{P}(W_2|U_1) \mathbb{P}(U_1) + \mathbb{P}(W_2|U_2) \mathbb{P}(U_2) \\ &= \frac{\binom{2}{2}}{\binom{6}{2}} \frac{1}{2} + \frac{\binom{3}{2}}{\binom{5}{2}} \frac{1}{2} = \frac{1}{15} \frac{1}{2} + \frac{3}{10} \frac{1}{2} = \frac{11}{60}. \end{aligned}$$

Now we can apply Bayes' rule:

$$\mathbb{P}(U_1|W_2) = \frac{\mathbb{P}(W_2|U_1) \mathbb{P}(U_1)}{\mathbb{P}(W_2)} = \frac{\frac{1}{30}}{\frac{11}{60}} = \frac{2}{11}.$$

SOME USEFUL TABLES AND FORMULAS

Shifting of ordinary generating functions:

If $A(z) = \sum_{n=0}^{\infty} a_n z^n$, then $\frac{A(z) - A(0)}{z} = \frac{A(z) - a_0}{z}$ is the OGF of the sequence $\langle a_{n+1} \rangle_{n=0}^{\infty}$.

$\frac{A(z) - a_0 - a_1 z}{z^2}$ is the OGF of the sequence $\langle a_{n+2} \rangle_{n=0}^{\infty}$.

Table of ordinary generating functions:

Function	Sequence	Series
$\frac{1}{1-z}$	$\langle 1, 1, 1, \dots \rangle$	$\sum_{n=0}^{\infty} z^n$
$\frac{1}{1+z}$	$\langle 1, -1, 1, -1, \dots \rangle$	$\sum_{n=0}^{\infty} (-1)^n z^n$
$\frac{1}{1-az}$	$\langle 1, a, a^2, a^3, \dots \rangle$	$\sum_{n=0}^{\infty} a^n z^n$
$\frac{1}{1-z^2}$	$\langle 1, 0, 1, 0, 1, \dots \rangle$	$\sum_{n=0}^{\infty} z^{2n}$
$\frac{1}{(1-z)^2}$	$\langle 1, 2, 3, 4, 5, \dots \rangle$	$\sum_{n=0}^{\infty} (n+1) z^n$

Formula for the total probability:

$$\mathbb{P}(B) = \sum_{i \geq 1} \mathbb{P}(B \cap A_i) = \sum_{i \geq 1} \mathbb{P}(B|A_i) \mathbb{P}(A_i)$$

Bayes' formula:
$$\mathbb{P}(A_k|B) = \frac{\mathbb{P}(B|A_k) \mathbb{P}(A_k)}{\sum_{i \geq 1} \mathbb{P}(B|A_i) \mathbb{P}(A_i)}$$